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Roll No. | | | | | | | | | | | | | **Total No. of Pages : 02**

Total No. of Questions : 18

B.Tech. (EE) PT (Sem.–1)

CIRCUIT THEORY

Subject Code : BTEE-301

M.Code : 70971

Time : 3 Hrs. **Max. Marks : 60**

INSTRUCTIONS TO CANDIDATES :

1. **SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.**
2. **SECTION - B & C. have FOUR questions each.**
3. **Attempt any FIVE questions from SECTION B & C carrying EIGHT marks each.**
4. **Select atleast TWO questions from SECTION - B & C.**

SECTION-A

Write Briefly / Fill in the blanks :

1. Differentiate dependent and independent sources.

2. State Superposition Theorem.

3. Distinguish between mesh and loop of an electric circuit.

4. Define quality factor of a series resonant circuit.

5. If a unit step current is passed through a capacitor, what will be the voltage across the capacitor?

6. Explain Convolution Theorem.

7. Define a transfer function.

8. Discuss the application of impedance and admittance parameters.

9. The cut-off frequency of constant k-low pass filter is_____.

10. The network function N (S) becomes _____ when s is equal to anyone of the zeros.

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SECTION-B

11. Determine R_L so as to have maximum power transfer to R_L in the given circuit.

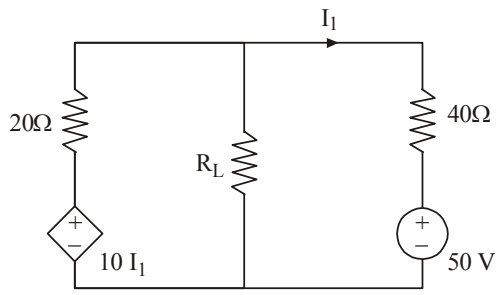

Fig. 1

12. Explain the initial value theorem.

13. Discuss the impedance parameters of a two port network.

14. Discuss the classification of filters.

SECTION-C

15. When is a network either T or π, is said to be of the constant-k type?

16. Discuss the response of the given RLC circuit excited by DC supply.

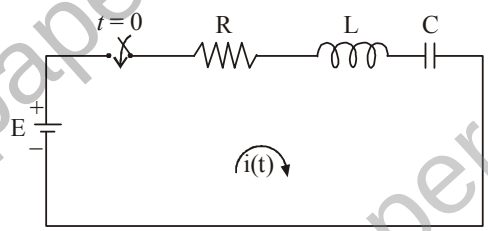

Fig. 2

17. Using Foster Form II synthesize the following function :

$$Z(s) = \frac{(s^2+5)\,(s^2+13)}{s(s^2+9)}$$

18. Write short notes :

    a) *m*-derived filters

    b) Cauer Forms.

NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.

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